

# Cooperative Air-Ground Instant Delivery by UAVs and Crowdsourced Taxis

Junhui Gao<sup>1</sup>, Qianru Wang<sup>2</sup>, Xin Zhang<sup>3</sup>, Juan Shi<sup>4</sup>, Xiang Zhao<sup>5</sup>, Qingye Han<sup>6</sup> and Yan Pan<sup>3,7\*</sup>

<sup>1</sup>School of Computer Science, Northwestern Polytechnical University,

<sup>2</sup>School of Computer Science and Technology, Xidian University,

<sup>3</sup> National Key Laboratory of Information Systems Engineering, National University of Defense Technology,

<sup>4</sup>Air Force Engineering University, <sup>5</sup> Laboratory for Big Data and Decision, National University of Defense Technology,

<sup>6</sup>School of Management Science and Real Estate, Chongqing University,

<sup>7</sup> National Key Laboratory of Parallel and Distributed Computing, National University of Defense Technology.

Emails: gaojh@mail.nwpu.edu.cn, wangqianru@xidian.edu.cn, shinezhang\_nudt@qq.com, shijuan-1029@163.com, xiangzhao@nudt.edu.cn, hanqy11@cqu.edu.cn, panyan@nudt.edu.cn.

**Abstract**—Instant delivery has become a fundamental service in people’s daily lives. Different from the traditional express service, the instant delivery has a strict shipping time constraint after being ordered. However, the labor shortage makes it challenging to realize efficient instant delivery. To tackle the problem, researchers have studied to introduce vehicles (i.e., taxis) or Unmanned Aerial Vehicles (UAVs or drones) into instant delivery tasks. Unfortunately, the delivery detour of taxis and the limited battery of UAVs make it hard to meet the rapidly increasing instant delivery demands. Under this circumstance, this paper proposes an air-ground cooperative instant delivery paradigm to maximize the delivery performance and meanwhile minimize the negative effects on the taxi passengers. Specifically, a data-driven delivery potential-demands-aware cooperative strategy is designed to improve the overall delivery performance of both UAVs and taxis as well as the taxi passengers’ experience. The experimental results show that the proposed method improves the delivery number by 30.1% and 114.5% compared to the taxi-based and UAV-based instant delivery respectively, and shortens the delivery time by 35.7% compared to the taxi-based instant delivery.

**Index Terms**—UAV, taxi, data-driven, instant delivery, cooperative air-ground delivery

## I. INTRODUCTION

Instant delivery, such as takeout food and emergency material (i.e., medicine and first-aid kits) supply, is essential to daily life, especially during the outbreak of Covid-19 [1] [2]. Because instant delivery bears the overwhelming advantage of being extremely fast (e.g., within one or a half hours) compared with traditional express services which take days. In the future, increasing immediate requirements will stimulate a dramatic number of instant deliveries. Specifically, a report on the global takeout market [3] predicts that the scale of the takeout market will reach \$4,024 billion in 2032, with an annual growth of 8.3%. Additionally, another recent report demonstrates that the annual instant delivery number in China, one of the largest instant delivery markets, is predicted to increase from 27.9 billion in 2021 to 100 billion in 2026, with an annual increasing rate of 28.0% [4]. Obviously, the instant

delivery service will become one of the promising delivery paradigms in the near future.

However, the labor shortage in many countries is a major obstacle to meeting the rapidly increasing demands for both the traditional express service and the instant delivery service [5]. For example, the labor shortage in the U.S. Postal Service led to slow or no deliveries throughout Greater Cleveland, elsewhere in the Ohio state, and across the USA in September 2022 [6]. Besides, since 2021 in some large cities in both China [7] and USA [8], not only a large number of online instant food orders suffer an extra long delivery time, but also a large number of instant orders fail to be delivered, due to the instant delivery courier shortage.

To address the aforementioned issues, the academic and industrial communities have investigated various novel instant delivery methods, which bring taxis [9]–[11] and Unmanned Aerial Vehicles (UAVs or drones) [12]–[15] into the instant delivery. For example, taxis are widely crowdsourced to deliver instant parcels all around the world [16], and UAV has been commercially implemented in instant delivery in Shenzhen and Shanghai by Meituan [17], one of the biggest instant delivery companies in China. Crowdsourcing instant delivery [10] encourages taxis to participate in delivery when they are available. Since any taxi can participate in instant delivery, crowdsourcing instant delivery also has the advantage of a large candidate pool of taxis to meet the increasing delivery demands. On the other hand, crowdsourcing instant delivery costs much less than traditional delivery because the company saves the cost of delivery trucks and full-time labor. Some works employ UAVs [12]–[15] as the emerging and promising delivery instrument for business use, which have the advantage of flexibility, faster delivery speed, less energy consumption, and less air pollution emission over ground vehicles [17].

However, the inherent limitations of the crowdsourced taxis and UAVs make them hard to deliver the instant parcels effectively and efficiently on their own. Specifically, the general drawback of the crowdsourced taxis is that the delivery capacity is directly determined by the passenger routes [10] [18] [19]. The different spatial-temporal distributions of the

\*Corresponding author: Yan Pan

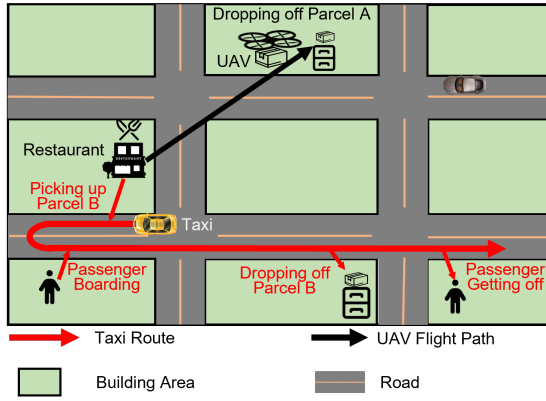


Fig. 1. The cooperative delivery by UAV and Taxi.

delivery orders and the taxi passengers will significantly impact and limit the delivery capacity, because there may not be enough available taxis near some delivery demands. Therefore, it is not efficient to solely use taxis for instant delivery. Though UAVs can quickly deliver an instant parcel, UAVs also have two main limitations: 1) Due to the energy storage technology, UAVs are restricted to a limited deliverable payload and deliverable range. 2) Another limitation is the lack of a mature and efficient low-altitude airspace traffic management system, so the UAVs have to keep an ultra-wide distance from each other to maintain flying safety. Thus, the capacity of UAVs for crowdsourcing tasks is dramatically limited, especially in the areas with a dense distribution of delivery tasks (e.g., the CBDs).

To fully leverage the advantages of both taxis and UAVs and avoid their drawbacks, this paper proposes an air-ground cooperative instant delivery paradigm to combine UAVs with taxis for more efficient instant deliveries. Specifically, the taxis only deliver the instant parcels close to the passenger routes, while other parcels far away from passengers are scheduled to be delivered by the UAVs. On one hand, the UAV instant delivery makes up for the inefficiency of the taxi-based instant delivery; On the other hand, the taxi-based delivery can ship the instant deliveries in densely distributed areas as well as deliver instant parcels in a wider space. The cooperative delivery scenario is illustrated in Fig.1. The taxi receives a passenger request from the origin to a specific destination. Meanwhile, two parcels (parcel A and parcel B in Fig.1) are ordered to two different destinations. Parcel A's destination is far away from the taxi routes, while Parcel B's destination is near the taxi route. Thus, Parcel A will be delivered by a UAV, and Parcel B can hitchhike the taxi to its destination [10]. The UAV flies from the restaurant to the destination of parcel A directly, the taxi picks up parcel B before picking up the passenger. Due to the short distance from Parcel B's destination to the passenger route, the taxi drops off Parcel B in halfway through the passenger route. Finally, the taxi arrives at the passenger's destination. The taxi delivery in this paradigm is different from the conventional taxi ride-sharing. That is a taxi can deliver different passengers with the same priority in the conventional ride-sharing problem [20], while the instant parcels and passengers have different

priorities in this study. Moreover, the taxi delivery capacity is affected by the UAV delivery capacity. Therefore, the main technical problem tackled in this paper is how to dispatch the parcels to achieve higher delivery performance with as little negative impact on the passenger experience as possible. The contributions of this paper are 3-fold:

- To the best of the authors' knowledge, this is the first work to study the innovative cooperative air-ground instant delivery by UAVs and crowdsourced taxis to make up for the inefficiency of sole taxi-based delivery and UAV-based delivery.
- This paper proposes a data-driven cooperative air-ground instant delivery framework to ship the instant deliveries efficiently. Specifically, a delivery potential-demands-aware cooperative strategy is designed to improve the overall delivery performance of both the UAVs and the taxis as well as the taxi passengers' experience.
- Using the large-scale, real-world instant delivery dataset and taxi trajectory dataset, and a commercial UAV's energy consumption model, a comprehensive evaluation is conducted to verify the cooperative air-ground instant delivery framework. The results show that the proposed paradigm 1) improves the delivered parcel number by 30.1% and 114.5% compared to the taxi-based and UAV-based instant delivery respectively, 2) shortens the delivery time by 35.7% compared to the taxi-based instant delivery method.

The remaining part of the paper is organized as follows. Sec.II reviews the existing works closely related to the air-ground instant delivery. Sec.III conducts a data-driven analysis of the research motivation of the work. The detailed design of the cooperative instant delivery model is introduced in Sec.IV. A delivery demand prediction model is utilized to guide the UAV repositioning, and finally, the optimization problem is formulated and solved in Sec.V. Sec.VI conducts a data-driven evaluation of the cooperative instant delivery framework, followed by the conclusion in Sec.VII.

## II. LITERATURE REVIEW

The instant delivery is the prevalent logistics service and the corresponding delivery dispatching problem has attracted the rising attention recently [21]–[24]. This section mainly reviews the related work on the instant delivery based on crowdsourced vehicles and UAVs, as well as other forms of air-ground delivery.

### A. Crowdsourcing Delivery

Wang et al. [9] designed a crowdsourcing instant delivery allocation strategy by considering the preference of crowdsourced social vehicles, so that the vehicle drivers can choose their most appropriate delivery orders to optimize the experience of both customers and passengers. Ding et al. [11] studied a crowdsourcing instant delivery system based on public transportation vehicles, and designed a delivery dispatching algorithm considering the time constraints, multi-hop relays, and delivery profits. Liu et al. [10] studied an

instant delivery network using the crowdsourced urban taxis. This work studied two forms of crowdsourcing instant delivery. One is that food is opportunistically delivered by taxis carrying passengers, and the other is that taxis dedicatedly deliver food without taking passengers. To address the labor shortage problem of crowdsourcing, reference [25] proposed that multiple delivery corporations can borrow unoccupied workers from each other to improve the quality of delivery service. Bai et al. [26] proposed a dynamic delivery pricing scheme to balance the delivery capacity of crowdsourcing instant delivery. Chen et al. [27] conducted a data-driven measurement of the instant delivery capacity of crowdsourced social vehicles. They found that the 14,000 taxis in Chengdu, one megacity in China, can deliver only about 10,000-20,000 parcels, due to the spatial-temporal diversity and dynamics of the passenger trips. Xu et al. [28] proposed to coordinate the dedicated couriers and crowdsourced taxis for food delivery.

### B. UAV Based Delivery

Kessens et al. [29] and Gawel et al. [12] designed aerial grasping technologies used by UAVs. With these technologies, UAVs can pick up and deliver items autonomously. Dissanayaka et al. [13] reviewed the navigation technologies needed by UAVs for delivery including sensor configurations, multi-sensor data fusion architectures, and their performance when employed for UAV navigation.

Another line of research is the scheduling problems of UAV-based delivery. Chauhan et al. [15] and Salama et al. [30] studied the joint station location determination and delivery planning to maximize the coverage of UAV-based delivery, limited by the available UAV and station number. Li et al. [14] investigated a comprehensive management framework for UAV-based delivery in the urban low-altitude space. The framework includes delivery path planning, aerial traffic management with conflict detection and resolution, the delivery resource allocation in conjunction with the payment scheme.

### C. Interactive Air-Ground Delivery

Cooperative air-ground delivery can effectively take advantage of both UAVs and vehicles and avoid their disadvantages. Existing works mainly focus on the interactive UAV-Vehicle delivery, in which the UAV rides the vehicle and takes the vehicle as a mobile launching and retrieval site [31] [32] to save energy. In these researches, the vehicle generally refers to a truck. Das et al. [33] studied the cooperative path planning of UAVs and vehicles to synchronize the both. Carlsson et al. [34] studied the delivery time minimization problem of the interactive UAV-Vehicle delivery problem. A heuristic path planning algorithm for both the UAVs and ground vehicles was proposed, whose approximation is theoretically proven. Wu et al. [2] proposed that the UAV-Vehicle delivery enabled contactless parcel delivery during the epidemic, and studied the routing problem. Except for trucks, public buses are also studied to be utilized to relay UAVs. For example, Pan et al. [32], [35] and Huang et al. [36] investigated to use the public buses to extend the delivery range of the UAVs.

TABLE I  
BASIC INFORMATION OF DATASETS

| Datasets        | # of couriers / taxis | Area( $km^2$ ) | # of Records | Sampling Interval |
|-----------------|-----------------------|----------------|--------------|-------------------|
| aBeacon         | $\geq 31K$            | 6400           | $\geq 802K$  | 1 – 3 min         |
| Taxi Trajectory | $\geq 13K$            | 6400           | $\geq 3.4B$  | 10 s              |
| Context-weather | N/A                   | 6400           | $\geq 8K$    | 30 min            |
| Context-region  | N/A                   | 6400           | $\geq 180K$  | N/A               |

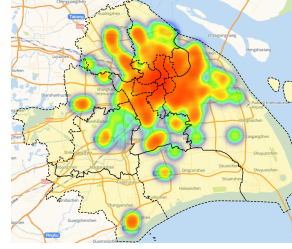


Fig. 2. Spatial Distribution of Pick-ups of Deliveries

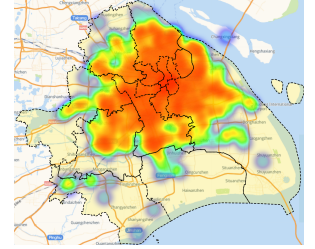


Fig. 3. Spatial Distribution of Pick-ups of Taxi Passengers

During the delivery using the interactive air-ground delivery, UAVs frequently interact (including land, take off, and plan path) with the ground vehicle. This interaction brings ultra-high system complexity and operation risk. So, there will be a long way to the wide application of the interactive air-ground delivery. The air-ground instant delivery strategy studied in this paper involves no direction UAV-Vehicle interaction, which is more feasible and practical.

## III. PRELIMINARY

### A. Datasets

Three real-world datasets respectively on instant deliveries, taxi trajectories, and contexts are used in this paper. The basic information of the datasets is shown in Table I.

**The aBeacon Dataset.** The one-month dataset is released by Yang et al. [37] and Ding et al. [38] cooperating with the Local Service Department of Alibaba Group, one of the world's largest instant delivery companies. The dataset includes over 800 thousand instant delivery orders ( $\approx 3000$  daily) delivered by 31,000 couriers at more than 2,400 merchant locations in the urban area of  $6400 km^2$  of Shanghai, China. Each order consists of the ordering time, the merchant location, the arrival time, and the customer location. Fig. 2 shows the spatial distribution of deliveries that occur between 12:00 and 12:30 in one day. The yellow-colored part is the area of Shanghai except for Chongming Island (the area is referred to Shanghai in this paper). Fig. 2 illustrates that most deliveries occur in the middle northern part (downtown area) during 12:00-12:30.

**The Taxi Trajectory Dataset.** This trajectory dataset of the taxi fleet in Shanghai is available at GitHub [39], which includes the moving status of over 13,000 taxis during 1 month. Each record consists of the timestamp, location, and passenger occupancy status, with a fine-grained sampling interval of about 10 seconds. The pick-ups of the taxi passengers from 12:00 to 12:30 are illustrated in Fig.3. The coverage average of

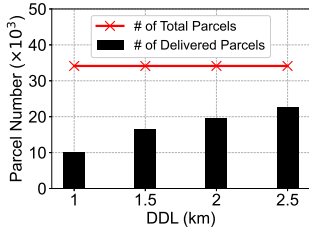


Fig. 4. Delivery Number by Different Detour Distance (DDL= 1.5km, PTR= 60%).



Fig. 5. Delivery # by Different Occurrence Time Limit (DDL= 1.5km, PTR= 60%).

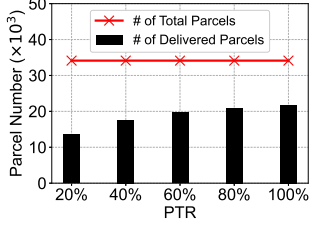


Fig. 6. Delivery Number by Different Participated Taxis (DDL= 1.5km, the Daytime (DDL= 1.5km, PTL= 3min).

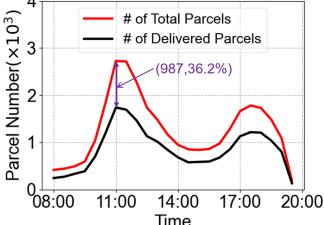


Fig. 7. Delivery Number during the Daytime (DDL= 1.5km, PTL= 3min, PTR= 100%).

the taxis is much wider than that of instant deliveries, showing the feasibility of taxi-based instant delivery. A deeper analysis of the spatial-temporal dynamics of both the passenger routes and the instant deliveries in the next subsection will reveal the limitation of taxi-based delivery, which motivates this work.

**The Contextual Dataset.** The contextual dataset consists of the weather condition dataset and the functionality area of the city. The sampling interval of weather conditions is 30min. There are over 1,400 weather condition records collected during the same period of the instant deliveries. Each weather condition record includes the weather, temperature, wind velocity, humidity, and air pressure, and the original data is collected from TimeAndDate [40]. Besides, the city is divided into 6400 grids and each grid is categorized into one of the 8 kinds of functionalities, including *Leisure*, *Commercial (Comm.)*, *Tourism*, *Healthcare (Health.)*, *Office*, *Residential (Resi.)*, *Study*, and *Industrial (Indust.)*. The functionality data is extracted from OpenStreetMap [41].

### B. Motivation

Some key definitions are introduced as follows.

**Definition III.1 (Detour Distance Limit, DDL).** DDL is the longest detour distance of the taxi being able to pick up the instant parcel, which significantly affects the passenger experience, and will be discussed in Sec.IV-C-ii).

**Definition III.2 (Pick-up Time Limit, PTL).** PTL is the time limitation from the instant parcel being ordered to the taxi being able to pick up the parcel. The pick-up time of the parcel by the taxi should hold Eq.(4) in Sec.IV-C-i).

**Definition III.3 (Participating Taxi Ratio, PTR).** Since only a part of the taxis are willing to deliver parcels, PTR denotes the ratio of taxis that are willing to deliver parcels:

$$PTR = \frac{\# \text{ of taxis willing for delivery}}{\# \text{ of all taxis}}. \quad (1)$$

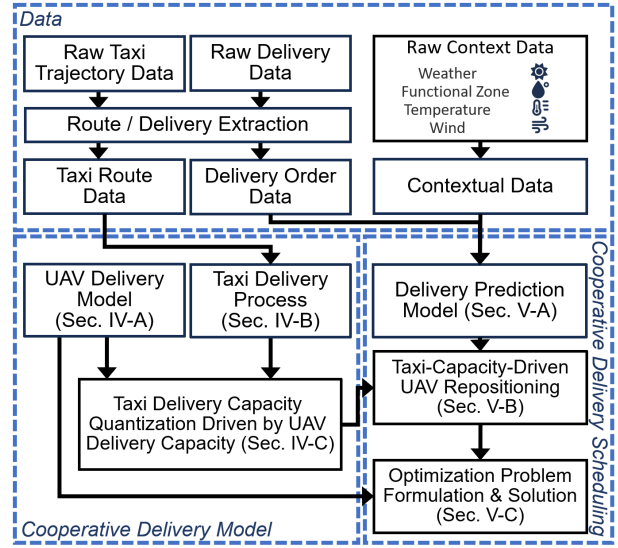


Fig. 8. The Aerial-Ground Instant Delivery Framework.

One taxi can deliver the instant parcels during a passenger route if they are spatially-temporally close. Specifically, the spatial limit refers to the origin and destination of a parcel should be both close enough to a taxi route. The close degree is pre-defined with DDL (e.g., 2km). The temporal constraint indicates that the taxi should pick up the parcel within a limited time range after the parcel order being released. The allowed time range is measured by PTL. Besides, PTR denotes the number of taxis that are willing to deliver parcels. Clearly, the DDL, PTL, and PTR have great impacts on the number of parcels that can be delivered by the taxis. To investigate the impacts of these factors, the delivery scheduling problem of assigning the taxis to deliver the instant parcels is formulated as the Generalized Assignment Problem (GAP). The heuristic scheduling algorithm for GAP [42], which minimizes the detour distance of each delivery by a taxi, is implemented. The influence of these factors on the Delivery Number by taxi is demonstrated in Fig.4, Fig.5, and Fig.6, respectively.

Fig.4 shows the delivered parcel number by taxis given different DDL. The values in the bracket in the figure caption are the default values of PTL and PTR. When the DDL increases from 1km to 2.5km, the delivery ratio by taxis only increases from 29.7% to 66.0%, namely a considerable number of parcels cannot be delivered by the taxis.

Fig.5 illustrates the impact of the PTL. Only 52.6% of instance parcels can be delivered by taxis when the PTL is 2min. When the PTL reaches 5min, 62.6% of parcels can be delivered by taxis, namely near 40% cannot be delivered within 5min after being ordered. The effect of participating taxi number (PTR) is shown in Fig.6. When the taxi number increases by 5× (PTR increases from 20% to 100%), the delivered parcel number by taxis only increases from 40% to 63.8%, and about 36.2% of instant parcels cannot be delivered by taxis. This is because of the well-known effect of marginal reward decreasing in crowdsourcing applications [43]–[45]. So, when some deliveries in the non-downtown areas are ordered, there may not be enough nearby taxis to deliver the

parcels, even with looser DDL and PTL.

Fig.7 demonstrates the number of parcels that can be delivered by taxis during different time periods in the daytime (from 08:00 to 20:00). It shows that even all the 13,000 taxis participate in delivery (PTR=100%), a considerable number of the instant parcels cannot be delivered by passenger routes. Specifically, 2725 deliveries (equal to 20% of the taxi number) are ordered from 11:00 to 11:30, while only 1738 parcels can be delivered by all the 13,000 taxis. 987 parcels, about 36.2% of all parcels, cannot hitchhike the taxis to their destinations.

In summary, the results demonstrate that the delivery capacity of crowdsourced taxis is dramatically limited, and extra instruments are needed for efficient instant delivery.

### C. Framework

This paper studies to cooperate UAVs [15] [17] [30] and crowdsourced taxis [10] to deliver the instant parcels efficiently. Therefore, a data-driven air-ground cooperative instant delivery framework is proposed, as shown in Fig. 8. The input of the framework is the taxi trajectory data, the instant delivery data, and the contextual data. Then, by combining the UAV delivery model (Sec.IV-A) and the data-driven taxi delivery process model (Sec.IV-B), the delivery capacity of taxis is quantitatively modeled (Sec.IV-C). By combining the taxi delivery capacity model (Sec.IV-C) and the instant delivery demand prediction results (Sec.V-A), a UAV repositioning scheme is designed to balance the delivery capability within different areas (Sec.V-B). Finally, a cooperative air-ground optimization problem is formulated and solved to assign the deliveries among UAVs and taxis (Sec.V-C).

## IV. COOPERATIVE DELIVERY MODEL

The delivery models of UAVs and taxis are respectively described in Sec.IV-A and Sec.IV-B. The delivery capacity of the taxi fleet is quantified in Sec.IV-C.

### A. UAV Delivery Model

The typical instant delivery process of a UAV consists of three steps [17] [46]:

- Step 1. The UAV departs from the charging station and picks up the parcel, with the time  $t_1(u, p)$ , where  $u$  and  $p$  are respectively the IDs of UAV and parcel;
- Step 2. The UAV delivers the parcel to the destination cabinet with the time  $t_2(u, p)$ ; and
- Step 3. The UAV returns to the charging station from the destination cabinet with the time  $t_3(u, p)$ .

Note that one UAV can start a new delivery during Step 3 if its remaining energy is sufficient, so the departing location in Step 1 may be not the charging station. The energy consumption of the UAV to deliver the parcel is:

$$\Delta E(u, p) = \beta(0)t_1(u, p) + \beta(p)t_2(u, p) + \beta(0)t_3(u, p), \quad (2)$$

where the  $\beta(0)$  is the energy consumption rate of  $u$  in Steps 1 and 3 (without carrying a parcel), and  $\beta(p)$  is the energy consumption rate in Step 2 (with carrying  $p$ ). The daytime (from 08:00 to 20:00), when the UAVs deliver the instant parcels, is divided into a series of time slots  $\mathbb{T} = \{t_0, t_1, \dots, t_n\}$ , each has the same duration, e.g., 1min. Let  $E(u, t)$  denote

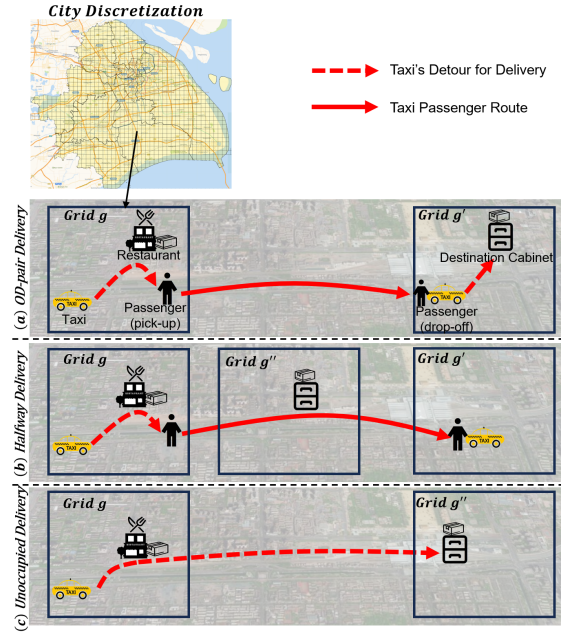


Fig. 9. Three Cases for Taxis Delivering Parcels.

the remaining energy of a UAV at the beginning of the time slot  $t \in \mathbb{T}$ . A binary notation is defined to indicate whether  $p$  can be delivered by  $u$ :

$$s(u, p) = \begin{cases} 1, & E(u, t) \geq \Delta E(u, p) \\ 0, & \text{otherwise} \end{cases}, \quad (3)$$

where  $s(u, p) = 1$  indicates the parcel  $p$  can be delivered by  $u$  because the UAV's energy  $E(u, t) \geq \Delta E(u, p)$  is sufficient.

### B. Taxis Delivery Process

The whole city is discretized into multiple grids  $\mathbb{G} = \{g_0, g_1, \dots, g_m\}$ , each with the same grid edge length (e.g., 1km). As illustrated in Fig.9, the delivery process of a taxi shipping an instant parcel can be discussed in 3 cases [18] [19] [10], based on the occupation status of the taxi and the spatial-temporal relationship of the taxi and the parcel. The red dotted line is the cruising trajectory of the taxi, the dash-dotted line represents the taxis' detour routes for parcel delivery, and the red curve is the taxi route for passenger transportation. Let  $\mathbb{D}(t, g)$  represent the set of the instant parcels whose origins are in the grid  $g$  during the time slot  $t$ .  $\mathbb{V}(t, g)$  and  $\mathbb{P}(t, g)$  respectively denote a set of taxis and passengers in  $g$  during  $t$ . Let  $\mathbb{R}(t, g, g')$  denote the set of passenger routes that head to the grid  $g'$  from  $g$  during  $t$ . The three taxi-based delivery cases are described as follows:

**Origin-Destination-pair (OD-pair) Delivery** [18]. As shown in Fig.9(a), the parcel to be delivered from  $g$  to  $g'$  can hitchhike the taxis in  $\mathbb{R}(t, g, g')$ . The taxi picks up the instant parcel from the parcel's origin (i.e., the restaurant) and then picks up the passenger. To maintain the passenger experience, the taxis will never pick up any parcel after picking up the passengers [18]. The taxi delivers the parcel after dropping off the passenger to guarantee the taxi passenger experience.

**Halfway Delivery** [19]. One instant parcel occurring in  $g$  can hitchhike a taxi carrying passengers from  $g$  to  $g'$  if the parcel's

destination cabinet (in grid  $g''$ ) is close to the passenger route halfway, as illustrated in Fig.9(b). Specifically, the taxi picks up the parcel before picking up the passenger and then drops off the parcel to the destination cabinet halfway through the passenger route.

**Unoccupied Delivery** [10]. There are some instant parcels that are destined for locations far away from the passenger routes (i.e., the grid  $g''$  in Fig.9(c)). Thus, the occupied taxis (namely carrying passengers) in  $\mathbb{R}(t, g, g')$  cannot deliver the parcels. Only the unoccupied taxis can detour to deliver these instant parcels, without passengers, as in Fig.9(c).

### C. Taxi Delivery Capacity Quantization

The delivery capacity of the taxis is numerically modeled based on OD-pair Delivery, Halfway Delivery, and Unoccupied Delivery of taxis, respectively.

**OD-pair Delivery Capacity.** One passenger route should satisfy the temporal-spatial limitations with a parcel  $p$  if the taxi  $v$  is able to participate in the OD-Pair Delivery,

- i) **The Temporal Limit.** The taxi  $v$  should arrive at the instant parcel  $p$ 's origin within the PTL (denoted as  $t_{max}$ ) after the order time of  $p$ . Then limitation of the pick-up time (denoted as  $t_{pu}(v, p)$ ) by  $v$  should satisfy:

$$t_{pu}(v, p) \leq t + \frac{|\mathbb{D}_{ud}(t, g)|}{|\mathbb{D}(t, g)|} \times t_{max}, \quad (4)$$

where  $t$  is the current time slot, the  $\mathbb{D}(t, g)$  and  $\mathbb{D}_{ud}(t, g)$  are the sets of the total instant parcels, and the undelivered instant parcels by UAVs during  $t$  at the grid  $g$ , respectively.  $\mathbb{D}_{ud}(t, g)$  is obtained by Eq.(3), and  $\frac{|\mathbb{D}_{ud}(t, g)|}{|\mathbb{D}(t, g)|} \leq 1$ . The temporal limitation used in the right expression in (4) is  $\frac{|\mathbb{D}_{ud}(t, g)|}{|\mathbb{D}(t, g)|} t_{max}$  instead of  $t_{max}$ . This is because if the UAVs deliver more instant parcels, there can be fewer taxis to participate in delivery.  $|\mathbb{D}_{ud}(t, g)|$  (and also  $\frac{|\mathbb{D}_{ud}(t, g)|}{|\mathbb{D}(t, g)|} t_{max}$ ) will be smaller. Thereafter, the exact temporal limit would be stricter, the average impact on the taxi passenger would be less, and vice versa.

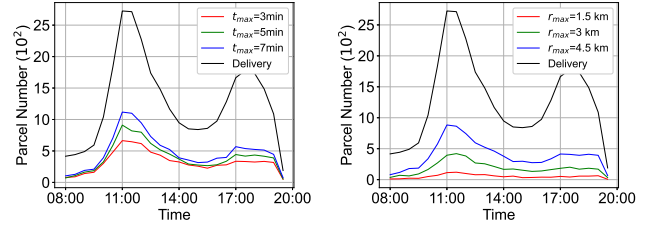
- ii) **The Spatial Limit.** The spatial limit in *OD-pair Delivery* is the detour for picking up instant parcels. Thus,  $p$  delivered by  $v$  in terms of the spatial limit should satisfy

$$r_{pu}(v, p) \leq \frac{|\mathbb{D}_{ud}(t, g)|}{|\mathbb{D}(t, g)|} \times r_{max}, \quad (5)$$

where  $r_{pu}(v, p)$  is the detour distances of  $v$  for picking up  $p$ ;  $r_{max}$  is the value of DDL. Similarly, considering the delivery of UAVs, the exact spatial limit of taxis is  $\frac{|\mathbb{D}_{ud}(t, g)|}{|\mathbb{D}(t, g)|} r_{max}$  instead  $r_{max}$ . Namely, if the UAVs deliver more parcels,  $\frac{|\mathbb{D}_{ud}(t, g)|}{|\mathbb{D}(t, g)|} r_{max}$  would be smaller, and the spatial limit (5) would be stricter, thus fewer taxis can participate in delivery.

The instant parcel  $p$  originating from grid  $g$  can be delivered by  $v$  in *OD-pair delivery* only if Eq.(4) and (5) are true.

**Halfway Delivery Capacity.** The temporal and spatial limits in *Halfway delivery* are the same as these in *OD-pair Delivery*. Because the passengers occupying the taxis enjoy the highest transportation priority, the passengers should



(a) Impact of PTL

(b) Impact of DDL

Fig. 10. Impact of Different Quantization Limits

be transported to their destinations within a distance limit [19], which is denoted as  $\Delta r$ . The detour of the taxi  $v$  for dropping off the instant parcel  $p$  halfway on the passenger route  $r_{do}(v, p)$  should satisfy

$$r_{do}(v, p) \leq \Delta r. \quad (6)$$

The instant parcel  $p$  which meets Eq.(4), (5), and (6) can be delivered by the taxi  $v$  in *Halfway Delivery*.

**Unoccupied Delivery Capacity.** When the parcels are delivered by the unoccupied taxis, the long delivery time and detour distance may also affect the potential passengers in the original passenger routes. Thus, the parcel delivery is also limited by the temporal and spatial features. The temporal limit is that the parcels must be picked up within PTL (Eq.4), while the spatial limit is the DDL, namely Eq.(5). Thus, the instant parcels can be delivered in *Unoccupied Delivery* if there are unoccupied taxis, and the Eq.(4) and the Eq.(5) are true.

**Summary.** The statuses of the instant parcels can be derived as follows.  $s(v, p) = 1$ , if the instant parcel  $p$  can be delivered by taxi  $v$  during the current time slot; otherwise,  $s(v, p) = 0$ .

$$s(v, p) = \begin{cases} 1, & ((4), (5)) \text{ or } ((4), (5), (6)) \text{ or } ((4), (5), |\mathbb{V}(t, g)| > |\mathbb{P}(t, g)|) \\ 0, & \text{otherwise} \end{cases} \quad (7)$$

where  $((4), (5))$  indicates the instant parcel  $p$  can be delivered by the taxi  $v$  in OD-pair;  $((4), (5), (6))$  represents  $p$  can be delivered halfway in the passenger route;  $((5), |\mathbb{V}(t, g)| > |\mathbb{P}(t, g)|)$  indicates  $p$  can be delivered by the unoccupied taxi  $v$ . The set of the instant parcels delivered by taxis can be derived from  $\mathbb{D}(t, g)$  by Eq.(7).

Fig.10(a) and Fig.10(b) show the impact of different values of the PTL and the DDL, respectively. The x-axes are the daytime with the interval of 30min, and the y-axes are the instant parcel number. Half of all taxis participate in delivery in the two figures. The black lines are the total instant parcel number, while the red, green, and blue lines indicate the number of the instant parcels that can be delivered by taxis with different quantization limits. The default DDL is 4.5km, while the default PTL is 5min. Fig.10(a) shows that the shorter PTL leads to fewer instant parcels delivered by taxis. This is because fewer taxis are available for the instant parcels in terms of the temporal domain. Over 1,100 of the instant parcels can be delivered from 11:00 to 11:30 with 7min of the PTL, which is intolerable for taxi passengers. However, only 664 of the instant parcels can be picked up during the same period of 3min after their occurrences. Besides, there

are two peaks of the instant parcel occurring, the noon peak (from 10:30 to 13:00) and the evening peak (from 17:00 to 18:30). The gaps of delivery capacity during the two peaks are significantly huge, about 59.2% and 68.5% of the instant parcels cannot be picked up within 7min. As shown in Fig.10(b), the shorter DDL leads to fewer instant parcels delivered by taxis. This is because fewer taxis are available for the instant parcels in terms of the spatial domain. Over 883 of the instant parcels are delivered by taxis from 11:00 to 11:30 with 4.5km of DDL, while the number of delivered instant parcels by taxis with 1.5km DDL is only 113 during the same period. Although the taxis can detour 4.5km to deliver the instant parcels, there are still a large number of undelivered instant parcels, especially in the noon peak (67.6%) and the evening peak (76.5%).

## V. COOPERATIVE DELIVERY SCHEDULING

This section first predicts the future delivery demands, then repositions the UAVs accordingly to fill the gap between the delivery demands and capacity. Finally, the scheduling optimization problem of the cooperative air-ground delivery is formulated and solved.

### A. Delivery Prediction Model

The historical delivery dataset and the contextual dataset (introduced in Sec.III-A) are used to train the prediction model. Several pioneering works propose that the weather condition has a great impact on the instant parcel orders [21], [47], [48]. The functionality of grids also affects the instant parcel order occurrence. As shown in Fig.11, the average number of the instant parcels occurring in different functionality grids. The *Office* grids have the most instant parcels occurring, about 22.3 per grid. The instant parcels seldom happen in *Industrial* grids, only 0.1 per grid. Besides, the number of the instant parcels happening in other kinds of grids also varies, which indicates that the instant parcels are also related to the grids' functionalities.

A Multiple-Layer Perceptron (MLP) with backpropagation is harnessed to predict the potential instant parcel orders. Fig.12 illustrates the structure of the MLP, whose layer number is  $M$ . The black lines show how the inputs get forward, and the red lines indicate the backpropagation of the gradients. The hidden layers in the MLP are all fully connected layers (FC Layers) with activation functions. Let  $q_1(i)$  and  $q_2(i)$  indicate the numbers of inputs and the neurons in the  $i$ -th ( $i \in [0, M]$ ) layer, respectively. The output of the  $i$ -th layer is

$$H_{out}(i) = \phi(H_{in}(i)W(i) + b(i)), \quad (8)$$

where  $H_{in}(i)$  is the input of this layer with the  $q_1$  features,  $W(i) \in \mathbb{R}^{q_1(i) \times q_2(i)}$  is the weight parameter,  $b(i) \in \mathbb{R}^{1 \times q_2(i)}$  is the offset parameter, and  $\phi()$  is the activation function. The Leaky ReLU is used as the activation function for all hidden

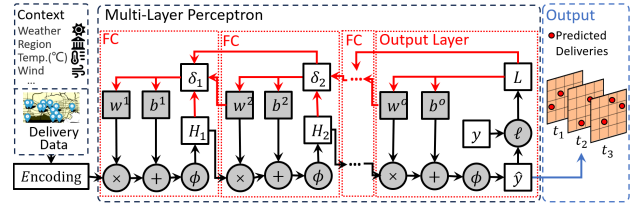


Fig. 12. The Structure of the Prediction Model.

layers, because of the positive inputs and the dying ReLU phenomenon [49]. Thus, the activation function output is:

$$\phi(z) = \begin{cases} z, & \text{if } z \geq 0 \\ 10^{-2}z, & \text{otherwise} \end{cases} \quad (9)$$

where  $z$  is the input of  $\phi()$ , and  $10^{-2}$  is the default value of the coefficient in Leaky ReLU in PyTorch [50].  $\hat{y}$  represents the output of the network. The predicted instant parcel number  $d(x)$  thus is

$$d(x) = \begin{cases} \lceil \hat{y} \rceil, & \text{if } \hat{y} \bmod 1 > 0.5 \\ \lfloor \hat{y} \rfloor, & \text{otherwise} \end{cases} \quad (10)$$

This work utilizes Huber Loss as the loss function, instead of the regularly used Mean Square Error (MSE) and Mean Absolute Error (MAE) [51]. This is because the MSE magnifies the error if there are outliers, which finally conducts a bad prediction; and the gradients of MAE always keep a higher value even for the small loss [52]. To enjoy the pros of MSE and MAE and avoid their cons, the Huber Loss measures the difference between the network output and the ground truth in the prediction model, which is both differentiable everywhere and robust to outliers [53], and provides a balance between accuracy and robustness [54]. The Huber Loss function is

$$L(y_i, \hat{y}_i) = \begin{cases} \frac{1}{2}(y_i - \hat{y}_i)^2, & |y_i - \hat{y}_i| \leq \delta \\ \delta|y_i - \hat{y}_i| - \frac{1}{2}\delta^2, & |y_i - \hat{y}_i| > \delta \end{cases}, \quad (11)$$

where  $\delta$  is the decision threshold for determining outliers [54] and set to the default value in PyTorch [55],  $y_i$  and  $\hat{y}_i$  are the ground truth and predicted label of the input, respectively. The obtained loss transmits backward along the red lines in Fig.12 and corrects the parameters in each layer. With the predicted  $d(x)$ , let  $\tilde{\mathbb{D}}(t, g)$  denote the predicted instant parcel orders in the grid  $g$  during the time slot  $t$ .

### B. Taxi-Capacity-Driven UAV Repositioning Model

Based on the predicted  $\tilde{\mathbb{D}}(t, g)$ , the UAVs are repositioned in advance to fulfill the delivery capacity gap in each grid. The unoccupied taxis and all passenger routes in the grid  $g$  during the time slot  $t$  are estimated based on the historical taxi trajectory data. Let  $\mathbb{G}(e)$  indicate the set of grids belonging to the UAV charging station  $e$ , while  $\mathbb{U}(e)$  is the set of the UAVs in  $e$ . Let  $s(v, :)$  denote whether the taxi  $v$  can deliver an instant parcel during  $t$  or not:  $s(v, :) = 1$  if  $s(v, p) = 1, \exists p \in \tilde{\mathbb{D}}(t, g)$ ; Otherwise,  $s(v, :)$  is 0. The number of instant parcels that can

be delivered by taxis in  $g$  during  $t$  is  $\sum_{v \in \mathbb{V}(t,g)} s(v, \cdot)$ . Thus, the delivery demand for the UAVs around  $e$  can be obtained:

$$\Delta C(e) = \sum_{g \in \mathbb{G}(e)} \sum_{t \in \Delta T} \left( \left| \tilde{\mathbb{D}}(t, g) \right| - \sum_{v \in \mathbb{V}(t, g)} s(v, \cdot) \right), \quad (12)$$

where  $\Delta T$  is the repositioning interval of the UAVs.  $\Delta C(e) < 0$  indicates that the taxis are abundant for delivery; while  $\Delta C(e) > 0$  indicates the UAVs have to participate in delivery.

To balance the delivery capacity among the charging stations and improve the delivery efficiency of UAVs, we reposition the UAVs from the stations that have 1) no delivery demand and 2) far delivery demands, to the stations that have closer delivery demands. A greedy algorithm is leveraged for repositioning: Firstly, the pick-up distances of all undelivered parcels by UAV  $u$  are calculated. If the parcel  $p$  is not around the charging station  $e$  where  $u$  belongs, the pick-up distance of  $p$  will be added with the detour distance to the station where  $p$  is around. Then,  $u$  is dispatched to deliver the parcel with the shortest pick-up distance and returns to the station of the parcel. This process is repeated until all parcels are delivered or no UAVs are available for delivery.

### C. Optimization Problem Formulation and Solution

With the models above, the air-ground cooperative delivery optimization problem is formulated as:

$$\max \sum_{g \in \mathbb{G}} \sum_{p \in \mathbb{D}(t, g)} \left( \sum_{v \in \mathbb{V}} \eta(v, p) + \sum_{u \in \mathbb{U}} \sigma(u, p) \right), \quad (13)$$

$$\min \sum_{g \in \mathbb{G}} \sum_{p \in \mathbb{D}(t, g)} \eta(v, p) r(v, p), \quad (14)$$

$$\text{s.t. } \eta(v, p) \in \{0, 1\}, \sigma(u, p) \in \{0, 1\} \quad (15)$$

$$s(v, p)\eta(v, p) + s(u, p)\sigma(u, p) \leq 1, \forall v \in \mathbb{V}, \forall u \in \mathbb{U}, \quad (16)$$

$$\sum_{g \in \mathbb{G}} \sum_{p \in \mathbb{D}(t, g)} s(v, p)\eta(v, p) \leq \alpha, \forall v \in \mathbb{V}, \quad (17)$$

$$\sum_{g \in \mathbb{G}} \sum_{p \in \mathbb{D}(t, g)} s(u, p)\sigma(u, p) \leq \gamma, \forall u \in \mathbb{U}. \quad (18)$$

The optimization problem is examined as follows. The first objective Eq.(13) is to maximize the total delivered parcel number by both the UAVs and taxis.  $t$  is the current time slot,  $\mathbb{G}$  is the grid set,  $\mathbb{D}(t, g)$  is the instant parcel set occurring in  $g$  during  $t$ ,  $\mathbb{V}$  and  $\mathbb{U}$  are respectively the taxi set and UAV set. The binary variables  $\eta(v, p)$  and  $\sigma(u, p)$  are the optimization objectives stated in (15). Specifically,  $\eta(v, p) = 1$  indicates that the parcel  $p$  is delivered by the taxi  $v$ , and  $\sigma(u, p) = 1$  indicates that  $p$  is delivered by the UAV  $u$ . The second objective (14) is to minimize the detour distance of the taxis transporting the passengers.  $s(v, p)$  and  $s(u, p)$  respectively represent the instant parcel  $p$  could be delivered by  $v$  and  $u$ , which are defined in Eq.(7) and Eq.(3). Eq.(16) represents that one instant parcel can only be delivered by one taxi  $v$  or one UAV  $u$ . Let  $\alpha$  represent the maximum delivery number of a

taxi simultaneously, which is called the delivery capacity of one taxi. Similarly, let  $\gamma$  denote the delivery capacity of a UAV. The constraint of the delivery capacity of a taxi and a UAV is stated in Eq.(17) and Eq.(18), respectively. This problem is a Generalized Assignment Problem with Assignment Restriction (GAPAR). A GAPAR instance refers to a set of bins (taxis and UAVs) and a set of items (parcels). Each bin has a capacity ( $\alpha$  for a taxi and  $\gamma$  for a UAV). Each assignment results in a profit (Eq.(13)) and cost (Eq.(14)). The approximate greedy algorithm [56] is utilized to solve the problem.

## VI. EVALUATION

### A. Evaluation Settings

1) *Datasets Division*: The one-month datasets introduced in Sec.III-A are used for evaluation. The data in the first 23 days is considered as the training dataset, and the data in the last 7 days is considered as the test dataset.

2) *Parameter Settings*: 6 important parameters that significantly influence the performance are considered in the model, i.e., delivery demand, the Participating Taxi Ratio (PTR), the number of UAVs in each charging station, the number of UAV charging stations, the Detour Distance Limit (DDL), and the Pick-up Time Limit (PTL). Specifically, the delivery demand refers to the portion of the instant parcels delivered by the cooperative delivery method. The delivery demand is in  $[0.6, 0.7, 0.8, 0.9, \underline{1}]$ , the PTR is in  $[5\%, \underline{10\%}, 15\%, 20\%]$ , the UAV number in each charging station is in  $[20, 25, \underline{30}, 35, 40]$ , the UAV charging station number is in  $[5, 10, \underline{15}, 20, 25]$ , the DDL is in  $[0.5km, 1, \underline{1.5}, 2]$ , and the PTL is in  $[3min, 4, \underline{5}, 6]$ . The underlined values are default.

The  $\alpha$  and  $\gamma$  are set to 1. The locations of UAV charging stations are randomly selected without conflicts in the downtown area of Shanghai. The UAV velocity, battery capacity, and flight

energy consumption rate ( $\beta$ ) refer to the characteristics of DJI M210 v2 [57]. Fig.13 illustrates the impacts of different parcel weights on the M210's energy consumption rate  $\beta$  (Watts). The blue dots are the energy consumption data of M210 from DJI [57], and the red curve indicates the linear fitting curve, whose fitting precision is very high with a minor error of only 2.4%. In the evaluation, the parcel weight is randomly chosen between 0.2kg to 1.0kg.

3) *Metrics*: 4 metrics are exploited to evaluate the delivery performance and the effect on the passengers of all algorithms.

- **Delivery Number** indicates the number of the instant parcels delivered by taxis and UAVs.
- **Delivery Time** is the time period from the origin of the instant parcel to its destination.
- **Passenger Extra Waiting Time (PEWT)** measures the passengers' experiences, which is caused by the taxis detouring for delivery.

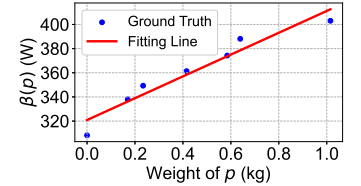


Fig. 13. The UAV Energy Consumption Rate Given Different Payload Weight.

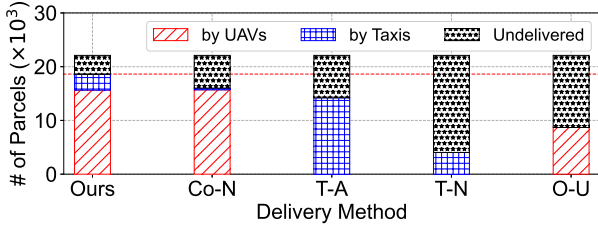


Fig. 14. # of the Instant Parcels Delivered by UAVs and Taxis

TABLE II  
RESULTS OF ABLATION STUDY

| Ablation       | Metrics | Delivery Number | Delivery Time (min) | PEWT (min) | UTPR (%) |
|----------------|---------|-----------------|---------------------|------------|----------|
| Ours           |         | 18,621          | 6.769               | 2.112      | 9.658    |
| T-C (Sec.IV-C) |         | 19,061          | 8.166               | 3.379      | 6.503    |
| D-P (Sec.V-A)  |         | 18,614          | 8.496               | 3.380      | 7.331    |
| U-R (Sec.V-B)  |         | 19,052          | 8.829               | 3.451      | 11.536   |

- **Unoccupied Taxi Participation Ratio (UTPR)** is the percentage of all unoccupied taxi routes that participate in instant parcel delivery.

4) *Baselines*: 4 baselines are implemented for comparison:

- **Parcels delivered by taxis (referred to T-A)** [10]. The method only employed taxis, including the taxis with passengers and the unoccupied ones to deliver the parcels.
- **Parcels delivered by taxi passenger routes (referred to T-N)**. Only the taxis with passengers are allowed to deliver the instant parcels in this method, due to the high cost of hiring the unoccupied taxis.
- **Parcels delivered by UAVs (referred to O-U)** [17] [46]. This method only allows UAVs to deliver instant parcels.
- **Parcels delivered by UAVs and the taxi passenger routes cooperatively (referred to Co-N)**. Taxi passenger routes and UAVs participate in the instant parcel delivery using the proposed cooperation delivery method. The unoccupied taxis are not allowed to participate in the instant parcel delivery in this method.

The proposed method is referred to **Ours** in the evaluation. The ablation study of **Ours** is first conducted in Sec.VI-B. The performance by **Ours** and the baselines under the default parameter settings are evaluated in Sec.VI-C, and the impacts of different parameters are evaluated from Sec.VI-D to Sec.VI-I. The experimental results shown in the following parts are the mean and Standard Error (SE) of results in 7 days.

### B. Ablation Study

We conduct an ablative study on each component of the framework as Fig.8, including 1) the Taxi Delivery Capacity Quantization Driven by UAV Delivery Capacity (Sec.IV-C) set to the fixed value, where  $\frac{|D_{ud}(t,g)|}{|D(t,g)|}$  in Eq.(4) and (5) is 1 (referred to **T-C**); 2) the current delivery orders rather delivery prediction (Sec.V-A) guide the UAV repositioning (referred to **D-P**); and 3) the UAVs will not be repositioned between charging stations (referred to **U-R**, Sec.V-B). The average results in 7 days are shown in Table II. Since more taxis will participate in delivery when the UAV delivery capacity is insufficient, and the delivery number in different methods stays stable. By dynamically adjusting the taxi delivery capacity by

UAVs, it is possible to ensure delivery performance during the rush hours and the passenger experience during the non-rush hours. If the taxi delivery capacity is fixed, the taxis with longer detours can pick up the parcels during the non-rush hours. It results in a much longer delivery time (20.6%) and PEWT (60.0%) than **Ours**, while the delivery number is comparable as in Table II. Delivery Prediction is the vital reference for the UAV repositioning, we can schedule the UAVs more accurately to the most needed charging stations, thus improving the delivery performance. The lack of the prediction may result in the wasted UAV delivery capacity, making more parcels delivered by taxis, finally prolonging the delivery time and PEWT by 25.5% and 60.0%, respectively. UAV repositioning schedules the UAVs between charging stations based on delivery efficiency. Without UAV repositioning, the UAV will always stay at the same charging station. Compared to the repositioning lack of delivery prediction, no UAV repositioning would increase the waste of UAV delivery capacity, resulting in the most unoccupied taxis (19.4% more than that of **Ours**) having to participate in parcel delivery.

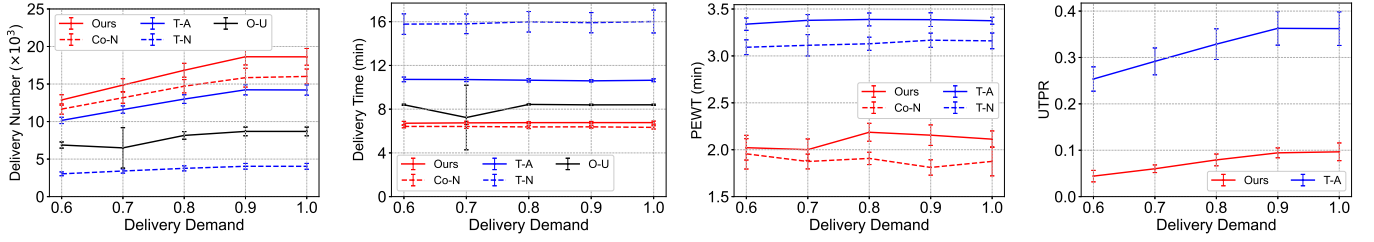
### C. Delivered Instant Parcel Number under Default Settings

Fig.14 illustrates the number of delivered instant parcels in different methods under the default parameter settings. 15,644 (70.8%) and 2,977 (13.5%) of the instant parcels are respectively delivered by UAVs and taxis, while only 3,480 (15.7%) instant parcels are not delivered on average in the proposed method. Due to the lack of unoccupied taxi delivery, the delivered instant parcel by taxis in **Co-N** is only 322, which is 89.2% less than that in **Ours**. Although 14,206 of the instant parcels are delivered in **T-A**, the total delivered parcel number is 23.7% less than the delivered parcel number in **Ours**. The taxis with passengers only deliver 4,037 (18.3%) of the instant parcels in **T-N**, which is only 21.7% of the delivered parcel number in **Ours**. Due to the flexibility of UAVs, the **O-U** achieves 13,416 (72.0% of delivered parcels in **Ours**) of the instant parcels delivered by only 450 UAVs.

### D. Impact of Delivery Demand

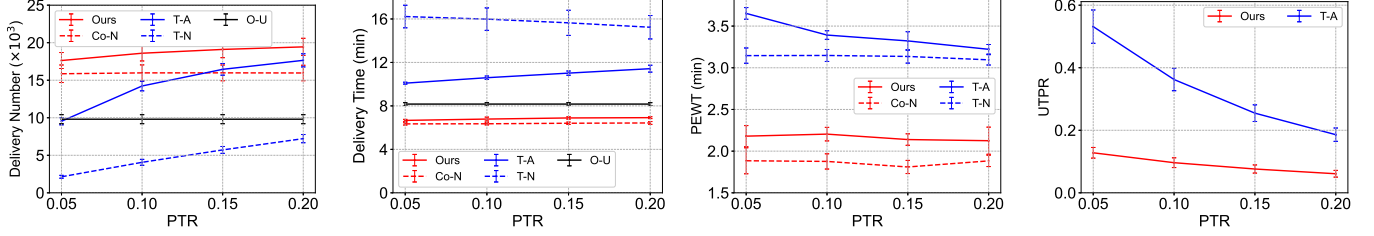
Fig. 15 shows the impact of delivery demand on the delivery performance and passenger experience.

Fig.15(a) shows the effect of the delivery demand on the delivery number. The delivered instant parcel number in **Ours** grows 44.7% when the delivery demand increases from 60% to 90%. This is because the higher delivery demand brings more instant parcels to be delivered. Besides, 18,633 of the instant parcels are delivered by **Ours** with 90% parcels available, which is 30.1%, 15.6%, and 114.5% more than delivered instant parcels in the **T-A**, the **Co-N** and the **O-U** baselines, respectively. This is because **Ours** exploits the UAVs, the taxis with passengers, and the unoccupied taxis participating in delivery. However, the delivery number stays stable when the delivery demand increases from 90% to 100%. This is because the delivery ranges of the taxis and UAVs are limited, there are thus no more instant parcels that can be delivered when all instant parcels are available. The increasing delivery demand has a slight impact on the delivery time as shown in



(a) Impact on Delivery Number (b) Impact on Delivery Time (c) Impact on PEWT (d) Impact on UTPR

Fig. 15. Impact of Delivery Demand on Delivery Performance and Passenger Experience



(a) Impact on Delivery Number (b) Impact on Delivery Time (c) Impact on PEWT (d) Impact on UTPR

Fig. 16. Impact of PTR on Delivery Performance and Passenger's Experience

Fig.15(b). The delivery time in all methods stays stable. This is because the parcels are evenly distributed across the city, the higher delivery demand does not change the distribution of the available parcels and the delivery constraints. The delivery time in **Ours** is about only  $6.7min$ , which is respectively 35.7%, 58.1% and 20.2% shorter than that in **T-A**, **T-N**, and **O-U**. This is because the UAVs and taxis can deliver the parcels that are not easy to deliver in the baselines while easy for them to deliver in **Ours**.

As a result of the increasing delivered parcel number, the passengers have to wait for a longer time (+4.5%), and more unoccupied taxis (+118.2%) participate in delivery services in **Ours**, as shown in Fig.15(c) and Fig.15(d). The taxis and UAVs prefer to deliver the closer instant parcels, which results in shorter PEWT as shown in 15(c). Besides, the passengers in **Ours** only need to wait for an extra  $2.1min$  to get on the taxi with all instant parcels available, which is 38.2% shorter than the time in **T-A**. Without UAVs, **T-A** has to dispatch the unoccupied taxis to deliver the instant parcels when the taxis with passengers are insufficient, which leads to the 275% higher UTPR than **Ours**. That is the other reason why the delivery time in **T-A** is 35.7% longer than in **Ours**. PEWT measures the taxi delivery cost, and the gain is the delivery performance including delivery number and delivery time. For passengers, **Ours** shortens the extra waiting time (PEWT) by 38.2% compared to **T-A**; for the delivery customers, **Ours** not only delivers more instant parcels (31.1%) but also achieves shorter delivery time (35.7%) than **T-A**.

#### E. Impact of PTR

Fig.16 plots the impacts of the Participating Taxi Ratio (PTR) on the delivery performance and passenger experience. This is because the PTR will not influence the UAV delivery, the curves of **UAV** stay stable.

Fig.16(a) shows the impact of the PTR on the delivery number. The results indicate that when more taxis participate

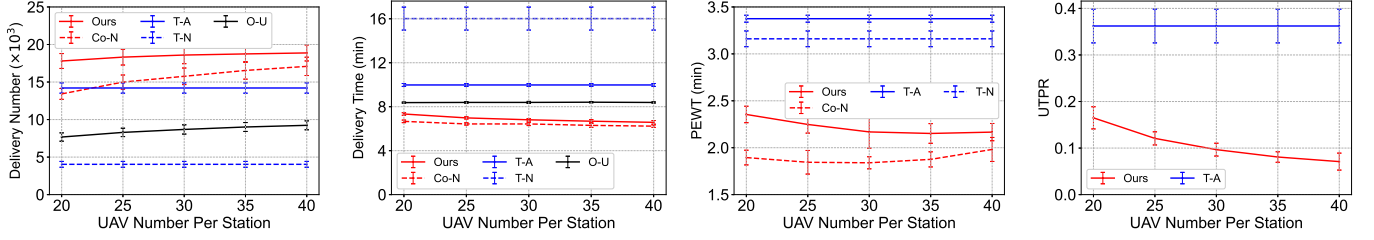
in the delivery service, the average delivered instant parcel number in **Ours** increases by 10.3%. When PTR is 0.2, over 19,400 of the instant parcels are delivered in **Ours**, which outperforms the baselines of **Co-N**, **T-A**, and **T-N** by 21.8%, 10.2%, and 169.3%, respectively. The impact on the delivery time is illustrated in Fig.16(b). Although the average delivery time in **Ours** increases  $0.3min$  (+4.5%), **Ours** still achieves the shortest delivery time among all methods.

With more available taxis, the parcels can hitchhike the passenger routes with shorter detours, which results in shorter PEWT in all methods as shown in Fig.16(c). The PEWT in **Ours** decreases only 2.4% with more taxis available. When the PTR is 0.2, the PEWT in **Ours** is only  $2.1min$ . This is because the UAVs can deliver most instant parcels in **Ours** and the more taxis participating in delivery will not significantly decrease the PEWT. Except for the PEWT, the UTPR also drops when PTR increases according to Fig.16(d). The UTPR in **Ours** drops 52.3%, and only 6.1% of all unoccupied taxis participating in delivery, which is 67.2% smaller than the UTPR in the **T-A**. This is because the UAVs can deliver the instant parcels when the taxis with passengers cannot.

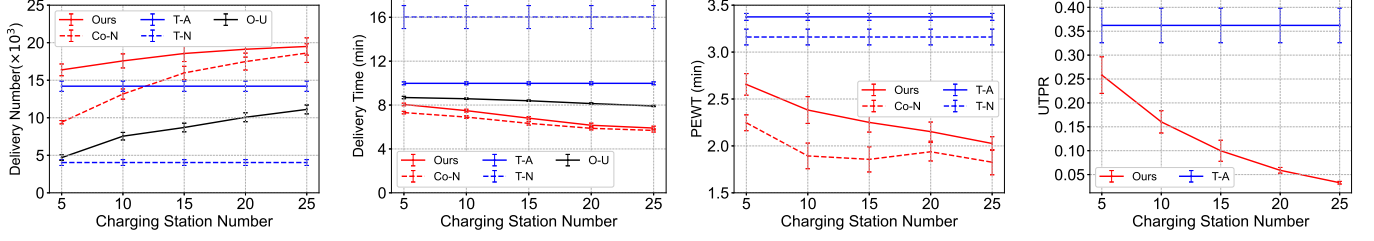
#### F. Impact of Number of UAVs in Each Charging Station

The influences of the UAV number in each station are shown in Fig.17. The curves of **T-A** and **T-N** remain stable.

More UAVs will deliver more instant parcels as shown in Fig.17(a). The number of delivered instant parcels in **Ours** grows only 6.0% when the number of UAVs in each charging station increases from 20 to 40, about 19,000 of the parcels are delivered with the most UAVs in the station. This is because when the UAV number per station is 20, the UAVs can deliver the vast majority of the instant parcels near the charging stations. The delivered parcel number does not increase significantly with more UAVs. **Ours** outperforms the baselines with all UAV numbers in each station. Especially,



(a) Impact on Delivery Number (b) Impact on Delivery Time (c) Impact on PEWT (d) Impact on UTPR  
Fig. 17. Impact of Number of UAVs in Each Charging Station on Delivery Performance and Passenger's Experience



(a) Impact on Delivery Number (b) Impact on Delivery Time (c) Impact on PEWT (d) Impact on UTPR  
Fig. 18. Impact of UAV Charging Station Number on Delivery Performance and Passenger's Experience

17,801 of the instant parcels are delivered in **Ours** when there are 20 UAVs in each station, which is 25.3% and 132.0% more than the delivered parcels in the **T-A** and **O-U** baselines, respectively. This is because **Ours** leverages the UAVs and all available taxis to deliver the instant parcels. Although more UAVs do not extend the delivery range of UAVs due to the limitations of the battery, the parcels will wait for a shorter time when the delivery demand is high. Thus, the increase of UAV number per station will shorten the delivery time of parcels as shown in Fig.17(b). The increase of UAV number shortens the delivery time by 11.2% in **Ours**. When the UAV number per station reaches 40, the delivery time in **Ours** is 21.6% shorter than the delivery time in **O-U**.

The more UAVs in each charging station also improves the passenger experience as illustrated in Fig.17(c). The passengers can wait for 2.17min to get on taxis when the UAV number per station is 40 in **Ours**, which is only 0.18min (+9.3%) longer than the PEWT in the **Co-N** baseline. As shown in Fig.17(d), the UTPR in **Ours** drops from 16.5% to 7.1%, which reveals the reason why PEWT drops. That is, the growth of the UAV delivery capacity leads to the less instant parcels delivered by unoccupied taxis. The passengers can thus wait for a shorter time to get on taxis.

#### G. Impact of UAV Charging Station Number

The impacts of the UAV charging station number are illustrated in Fig.18. The **T-A** and **T-N** curves are stable because the UAV station number is not involved in the two baselines.

As illustrated in Fig.18(a), the delivered instant parcel number in **Ours** grows 16.0% when the number of UAV charging stations grows from 5 to 25. Besides, most instant parcels are delivered in **Ours** as the number of UAV charging stations increases. Especially when there are only 5 stations, 16,379 of the instant parcels are delivered in **Ours**, which are 74.1% and 249.4% more than **Co-N** and **O-U**. Fig.18(b) plots the influences on the delivery time. The parcel delivery time

drops 36.4% with more UAV charging stations. This is because more charging stations enable UAVs to fly in a wider range in the city and deliver more parcels. With 5 UAV charging stations, the delivery time in **Ours** is 10.1% longer than that in **Co-N** and 7.3% shorter than that in **O-U**.

Fig.18(c) illustrates that the PEWT of **Ours** drops 0.63min (−23.7%) when the number of UAV charging stations grows. When there are only 5 charging stations, the passengers in **Ours** need to wait for extra 2.66min to get on taxis, which is only 0.41min (+18.2%) longer than the PEWT in **Co-N**. Fig.18(d) illustrates that the participating unoccupied taxis drop 87.2%. It indicates that the unoccupied taxis are replaced by UAVs for delivery when the UAV charging station number increases, which results in that the delivery number in **Ours** only increases by 18.3% as shown in Fig.18(a). The results show that **Ours** achieves considerable delivery performance when the UAV number is small, and a larger UAV number cannot improve the delivery performance dramatically.

#### H. Impact of DDL

The impacts of the DDL are depicted in Fig.19, in which the curves of **O-U** stay stable because DDL only affects the methods involving taxis.

Fig.19(a) depicts the delivered parcel number in **Ours** rises from 17,193 to 19,092 (+11.0%) when the DDL increases from 0.5km to 2km, because the longer DDL allows more taxis for delivery. With DDL= 2km, **Ours** delivers 19.8%, 19.9%, and 289.2% more parcels than **Co-N**, **T-A**, and **T-N**. This is because the UAVs can deliver the instant parcels which cannot be delivered by taxis. As more taxis participate in delivery, the delivery time in **Ours** rises 8.6% as shown in Fig.19(b). Compared with the baselines involving taxis, the delivery time in **Ours** is −9.2%, +38.9%, and 58.9% shorter than **Co-N**, **T-A**, and **T-N**, respectively. This is because the long detour distance parcels are delivered by UAVs in **Ours**.

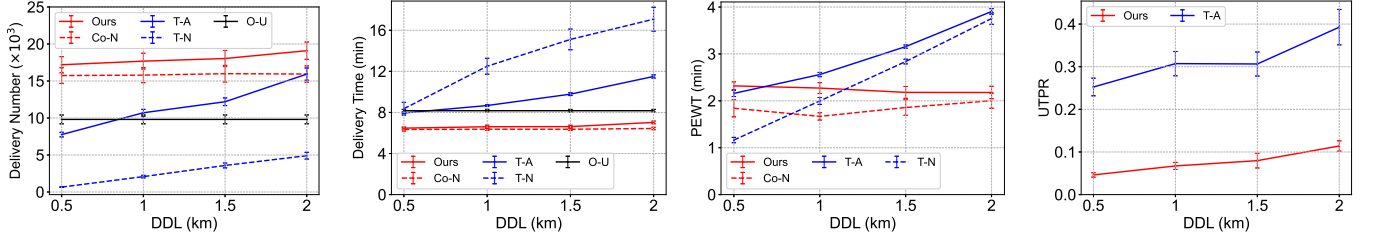


Fig. 19. Impact of DDL on Delivery Performance and Passenger's Experience

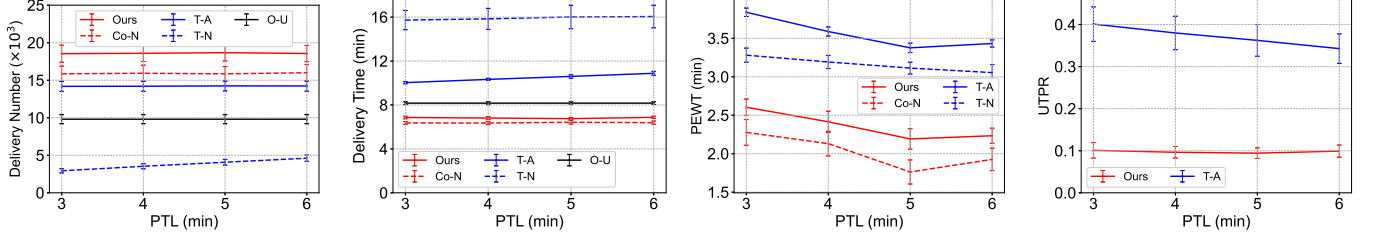


Fig. 20. Impact of PTL on Delivery Performance and Passenger's Experience

The PEWT changes with the DDL as shown in Fig.19(c). The PEWT in **Ours** slightly decreases from  $2.32min$  to  $2.18min$  ( $-6.4\%$ ). This is because the longer DDL allows the taxis further from the parcels to deliver them. Since the delivery capacity of taxis with passengers is limited by the UAVs, more unoccupied taxis participate in delivery. When the DDL is  $2km$ , the PEWT in **Ours** is  $-8.7\%$ ,  $44.23\%$  and  $42.0\%$  shorter than that in **Co-N**, **T-A** and **T-N**, respectively. Both the taxis with and without passengers can deliver further instant parcels when the DDL increases. This is the reason why the UTPR goes up from  $0.04$  to  $0.11$  ( $+1287.5\%$ ) when DDL increases from  $0.5km$  to  $2km$  in Fig.19(d). Besides, fewer unoccupied taxis participate in delivery ( $-71.0\%$ ) in **Ours** compared to **T-A** with  $DDL=2km$ .

#### I. Impact of PTL

Fig.20 illustrates the impacts of PTL on the delivery performance and passenger experience. The UAVs are not affected by PTL, thus the curves of **O-U** do not change in this figure.

Compared to **Co-N**, **T-A** and **T-N**, **Ours** delivers  $16.0\%$ ,  $30.4\%$ , and  $302.3\%$  more parcels respectively, despite a slight improvement in the number of instant deliveries when PTL ranges from  $3min$  to  $6min$ . This is because **Ours** uses three kinds of tools to deliver the instant parcels, the UAVs, the taxis with passengers, and the unoccupied taxis. Fig.20(b) displays the impacts of the PTL on delivery time. The delivery time of **Ours** will not increase significantly when PTL changes. Besides, the delivery time in **Ours** achieves the  $-7.6\%$ ,  $36.8\%$ , and  $57.2\%$  shorter delivery time compared with **Co-N**, **T-A**, and **T-N**, respectively. This is because the unoccupied taxis and the UAVs can directly deliver to the parcel destinations.

Fig.20(c) illustrates that the PEWT decreases  $14.2\%$  when the PTL increases from  $3min$  to  $6min$  in the proposed method. The PEWT in **Ours** is  $16.0\%$  longer than that in **Co-N** due to the participation of the unoccupied taxis, and is  $35.0\%$  shorter than the PEWT in **T-A** due to the participation

of the UAVs when the PTL is  $6min$ . According to Fig.20(d), the UTPR in **Ours** stays stable around  $0.1$ . This is because the most delivered parcels can be picked up within  $3min$  after their occurrences. Thus, the delivery number in **Ours** increases slightly (shown in Fig.20(a)). Besides, the UTPR in **T-A** is larger than that in **Ours**. **T-A** uses  $245.7\%$  more unoccupied taxis than **Ours** for delivery when the  $PTL=6min$ . Therefore, the passengers in **T-A** have to wait for a longer time than the passengers in **Ours** (shown in Fig.20(c)).

## VII. CONCLUSION

This paper proposes the first air-ground cooperative delivery framework for the instant delivery by UAVs and crowdsourced taxis based on the multi-source data, which aims to improve the delivery performance and maintain the taxi passengers' experience. The delivery models of UAVs and taxis are proposed, and then the UAVs' and taxis' delivery capacities are quantified. To improve the UAV delivery efficiency, they are repositioned among stations based on the delivery demand prediction. Thereafter, the air-ground cooperative delivery optimization problem is formulated and solved. Finally, the framework is evaluated with the real-world delivery dataset of 31,000 couriers and the taxi trajectory dataset of 13,000 taxis, which show that the proposed method improves the delivery number by  $30.1\%$  and  $114.5\%$  compared to the taxi-based and the UAV-based delivery respectively, and decreases the delivery time by  $35.7\%$  compared to the taxi-based delivery.

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## REFERENCES

- [1] C. Xiang, Y. Zhou, H. Dai, Y. Qu, S. He, C. Chen, and P. Yang, "Reusing delivery drones for urban crowdsensing," *IEEE Transactions on Mobile Computing*, 2021.
- [2] G. Wu, N. Mao, Q. Luo, B. Xu, J. Shi, and P. N. Suganthan, "Collaborative truck-drone routing for contactless parcel delivery during the epidemic," *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 12, pp. 25 077–25 091, 2022.
- [3] F. M. Insights. Restaurant takeout market outlook (2022–2032). [Online]. Available: <https://www.futuremarketinsights.com/reports/restaurant-takeout-market>
- [4] iresearch. China instant delivery industry trend research report in 2022. [Online]. Available: <https://report.iresearch.cn/report/202203/3964.shtml>
- [5] CNBC. There are millions of jobs, but a shortage of workers: Economists explain why that's worrying. [Online]. Available: <https://www.cnbc.com/2021/10/20/global-shortage-of-workers-whats-going-on-experts-explain.html>
- [6] R. Higgs. U.s. postal service struggles with worker shortage, delivery delays in greater cleveland, ohio and the nation. [Online]. Available: <https://www.cleveland.com/metro/2022/09/us-postal-service-struggles-with-shortage-of-workers-delays-in-delivery.html>
- [7] Y. Ao. (2022) Courier shortage: The insight of the poor and unfast delivery in multiple regions in china (in chinese). [Online]. Available: <https://new.qq.com/rain/a/20221222A0404T00>
- [8] V. CAMPISI. (2021) Labor shortage update: Restaurants limit delivery, online sales to focus on dine-in. [Online]. Available: <https://foodinstitute.com/focus-labor-shortage-update-restaurants-limit-delivery-online-sales-to-focus-on-dine-in/>
- [9] X. Wang, L. Wang, S. Wang, J. Pan, H. Ren, and J. Zheng, "Recommending-and-grabbing: A crowdsourcing-based order allocation pattern for on-demand food delivery," *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 1, pp. 838–853, 2022.
- [10] Y. Liu, B. Guo, C. Chen, H. Du, Z. Yu, D. Zhang, and H. Ma, "Foodnet: Toward an optimized food delivery network based on spatial crowdsourcing," *IEEE Transactions on Mobile Computing*, vol. 18, no. 6, pp. 1288–1301, 2019.
- [11] Y. Ding, B. Guo, L. Zheng, M. Lu, D. Zhang, S. Wang, S. H. Son, and T. He, "A city-wide crowdsourcing delivery system with reinforcement learning," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 5, no. 3, pp. 1–22, 2021.
- [12] A. Gawel, M. Kamel, T. Novkovic, J. Widauer, D. Schindler, B. P. Von Altshofen, R. Siegwart, and J. Nieto, "Aerial picking and delivery of magnetic objects with mav," in *2017 IEEE international conference on robotics and automation (ICRA)*. IEEE, 2017, pp. 5746–5752.
- [13] D. Dissanayaka, T. R. Wanasinghe, O. De Silva, A. Jayasiri, and G. K. Mann, "Review of navigation methods for uav-based parcel delivery," *IEEE Transactions on Automation Science and Engineering*, 2023.
- [14] A. Li, M. Hansen, and B. Zou, "Traffic management and resource allocation for uav-based parcel delivery in low-altitude urban space," *Transportation Research Part C: Emerging Technologies*, vol. 143, p. 103808, 2022.
- [15] D. Chauhan, A. Unnikrishnan, and M. Figliozzi, "Maximum coverage capacitated facility location problem with range constrained drones," *Transportation Research Part C: Emerging Technologies*, vol. 99, pp. 1–18, 2019.
- [16] R. T. Cab. Package delivery. [Online]. Available: <https://www.redtopcab.com/our-services/package-delivery/>
- [17] M. T. Review. Drone food delivery is now part of daily life in shenzhen. [Online]. Available: <https://www.technologyreview.com/2023/05/23/1073500/drone-food-delivery-shenzhen-meituan/>
- [18] C. Chen, D. Zhang, X. Ma, B. Guo, L. Wang, Y. Wang, and E. Sha, "crowddeliver: Planning city-wide package delivery paths leveraging the crowd of taxis," *IEEE Transactions on Intelligent Transportation Systems*, vol. 18, no. 6, pp. 1478–1496, 2017.
- [19] B. Li, D. Krushinsky, H. A. Reijers, and T. Van Woensel, "The share-a-ride problem: People and parcels sharing taxis," *European Journal of Operational Research*, vol. 238, no. 1, pp. 31–40, 2014.
- [20] U. D. of Transportation. Ride sharing programs. [Online]. Available: <https://www.transportation.gov/mission/health/ride-sharing-programs>
- [21] Y. Zhang, Y. Liu, G. Li, Y. Ding, N. Chen, H. Zhang, T. He, and D. Zhang, "Route prediction for instant delivery," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 3, no. 3, pp. 1–25, 2019.
- [22] Y. Yang, Y. Ding, D. Yuan, G. Wang, X. Xie, Y. Liu, T. He, and D. Zhang, "Transloc: transparent indoor localization with uncertain human participation for instant delivery," in *Proceedings of the 26th Annual International Conference on Mobile Computing and Networking*, 2020, pp. 1–14.
- [23] M. Joshi, A. Singh, S. Ranu, A. Bagchi, P. Karia, and P. Kala, "Batching and matching for food delivery in dynamic road networks," in *2021 IEEE 37th International Conference on Data Engineering (ICDE)*, 2021, pp. 2099–2104.
- [24] D. Chen, Y. Yuan, W. Du, Y. Cheng, and G. Wang, "Online route planning over time-dependent road networks," in *2021 IEEE 37th International Conference on Data Engineering (ICDE)*, 2021, pp. 325–335.
- [25] Y. Cheng, B. Li, X. Zhou, Y. Yuan, G. Wang, and L. Chen, "Real-time cross online matching in spatial crowdsourcing," in *2020 IEEE 36th International Conference on Data Engineering (ICDE)*, 2020, pp. 1–12.
- [26] S. Bai, S. Tong, X. Feng, Z. Jiang, X. Bai, and R. Xu, "Toward dynamic pricing for city-wide crowdsourced instant delivery services," *IEEE Transactions on Mobile Computing*, pp. 1–16, 2022.
- [27] Y. Chen, D. Guo, M. Xu, G. Tang, and G. Cheng, "Measuring maximum urban capacity of taxi-based logistics," *IEEE Transactions on Intelligent Transportation Systems*, vol. 22, no. 10, pp. 6449–6459, 2020.
- [28] X. Xu, A. Liu, G. Liu, Z. Li, and L. Zhao, "Drive less but finish more: Food delivery based on multi-level workers in spatial crowdsourcing," in *Proceedings of the 31st ACM International Conference on Information & Knowledge Management*, 2022, pp. 2331–2340.
- [29] C. C. Kessens, J. Thomas, J. P. Desai, and V. Kumar, "Versatile aerial grasping using self-sealing suction," in *2016 IEEE International Conference on Robotics and Automation (ICRA)*, 2016, pp. 3249–3254.
- [30] M. Salama and S. Srinivas, "Joint optimization of customer location clustering and drone-based routing for last-mile deliveries," *Transportation Research Part C: Emerging Technologies*, vol. 114, pp. 620–642, 2020.
- [31] Q. Luo, G. Wu, A. Trivedi, F. Hong, L. Wang, and D. Srinivasan, "Multi-objective optimization algorithm with adaptive resource allocation for truck-drone collaborative delivery and pick-up services," *IEEE Transactions on Intelligent Transportation Systems*, 2023.
- [32] Y. Pan, S. Li, Q. Chen, N. Zhang, T. Cheng, Z. Li, B. Guo, Q. Han, and T. Zhu, "Efficient schedule of energy-constrained uav using crowd-sourced buses in last-mile parcel delivery," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 5, no. 1, pp. 1–23, 2021.
- [33] D. N. Das, R. Sewani, J. Wang, and M. K. Tiwari, "Synchronized truck and drone routing in package delivery logistics," *IEEE Transactions on Intelligent Transportation Systems*, vol. 22, no. 9, pp. 5772–5782, 2020.
- [34] J. G. Carlsson and S. Song, "Coordinated logistics with a truck and a drone," *Management Science*, vol. 64, no. 9, pp. 4052–4069, 2018.
- [35] Y. Pan, Q. Chen, N. Zhang, Z. Li, T. Zhu, and Q. Han, "Extending delivery range and decelerating battery aging of logistics uavs using public buses," *IEEE Transactions on Mobile Computing*, 2022.
- [36] H. Huang, A. V. Savkin, and C. Huang, "Drone routing in a time-dependent network: Toward low-cost and large-range parcel delivery," *IEEE Transactions on Industrial Informatics*, vol. 17, no. 2, pp. 1526–1534, 2020.
- [37] Y. Ding, L. Liu, Y. Yang, Y. Liu, D. Zhang, and T. He, "From conception to retirement: A lifetime story of a 3-year-old wireless beacon system in the wild," *IEEE/ACM Transactions on Networking*, vol. 30, no. 1, pp. 47–61, 2022.
- [38] Y. Yang, Y. Ding, D. Yuan, G. Wang, X. Xie, Y. Liu, T. He, and D. Zhang, "Transloc: Transparent indoor localization with uncertain human participation for instant delivery," in *Proceedings of the 26th Annual International Conference on Mobile Computing and Networking*, ser. MobiCom '20. New York, NY, USA: Association for Computing Machinery, 2020.
- [39] cbdog94. Stl: Online detection of taxi trajectory anomaly based on spatial-temporal laws. [Online]. Available: <https://github.com/cbdog94/STL>
- [40] Time and Date. Weather in shanghai. [Online]. Available: <https://www.timeanddate.com/weather/china/shanghai>
- [41] O. S. Map. [Online]. Available: <https://www.openstreetmap.org/>

- [42] R. Cohen, L. Katzir, and D. Raz, "An efficient approximation for the generalized assignment problem," *Information Processing Letters*, vol. 100, no. 4, pp. 162–166, 2006.
- [43] Y. Pan, S. Li, Z. Ning, B. Li, Q. Zhang, and T. Zhu, "ausense: collaborative airspace sensing by commercial airplanes and unmanned aerial vehicles," *IEEE Transactions on Vehicular Technology*, vol. 69, no. 6, pp. 5995–6010, 2020.
- [44] M. Xiao, J. Wu, H. Huang, L. Huang, and C. Hu, "Deadline-sensitive user recruitment for mobile crowdsensing with probabilistic collaboration," in *2016 IEEE 24th International Conference on Network Protocols (ICNP)*. IEEE, 2016, pp. 1–10.
- [45] M. Xiao, J. Wu, L. Huang, Y. Wang, and C. Liu, "Multi-task assignment for crowdsensing in mobile social networks," in *2015 IEEE Conference on Computer Communications (INFOCOM)*. IEEE, 2015, pp. 2227–2235.
- [46] C. Xiang, Y. Zhou, H. Dai, Y. Qu, S. He, C. Chen, and P. Yang, "Reusing delivery drones for urban crowdsensing," *IEEE Transactions on Mobile Computing*, vol. 22, no. 5, pp. 2972–2988, 2023.
- [47] J. Liang, J. Ke, H. Wang, H. Ye, and J. Tang, "A poisson-based distribution learning framework for short-term prediction of food delivery demand ranges," *IEEE Transactions on Intelligent Transportation Systems*, pp. 1–14, 2023.
- [48] B. Guo, S. Wang, Y. Ding, G. Wang, S. He, D. Zhang, and T. He, "Concurrent order dispatch for instant delivery with time-constrained actor-critic reinforcement learning," in *2021 IEEE Real-Time Systems Symposium (RTSS)*, 2021, pp. 176–187.
- [49] L. Lu, Y. Shin, Y. Su, and G. E. Karniadakis, "Dying relu and initialization: Theory and numerical examples," *arXiv preprint arXiv:1903.06733*, 2019.
- [50] PyTorch. Leakyrelu-pytorch 2.1 documentation. [Online]. Available: <https://pytorch.org/docs/stable/generated/torch.nn.LeakyReLU.html>
- [51] T. Chai and R. R. Draxler, "Root mean square error (rmse) or mean absolute error (mae)? – arguments against avoiding rmse in the literature," *Geoscientific Model Development*, vol. 7, no. 3, pp. 1247–1250, 2014. [Online]. Available: <https://gmd.copernicus.org/articles/7/1247/2014/>
- [52] D. Chicco, M. J. Warrens, and G. Jurman, "The coefficient of determination r-squared is more informative than smape, mae, mape, mse and rmse in regression analysis evaluation," *PeerJ Computer Science*, vol. 7, p. e623, 2021.
- [53] G. P. Meyer, "An alternative probabilistic interpretation of the huber loss," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2021, pp. 5261–5269.
- [54] J. Ge, H. Li, H. Wang, H. Dong, H. Liu, W. Wang, Z. Yuan, J. Zhu, and H. Zhang, "Aeromagnetic compensation algorithm robust to outliers of magnetic sensor based on huber loss method," *IEEE Sensors Journal*, vol. 19, no. 14, pp. 5499–5505, 2019.
- [55] PyTorch. Huber loss-pytorch 2.1 documentation. [Online]. Available: <https://pytorch.org/docs/stable/generated/torch.nn.HuberLoss.html>
- [56] R. Cohen, L. Katzir, and D. Raz, "An efficient approximation for the generalized assignment problem," *Information Processing Letters*, vol. 100, no. 4, pp. 162–166, 2006. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0020019006001931>
- [57] DJI. Dji matrice 210 v2 features. [Online]. Available: <https://www.dji.com/cn/matrice-200-series-v2/info#specs>